Fused FP8 MoE Low-Latency Pipeline

Gather-free, PDL-chained FP8 experts for few-token decode

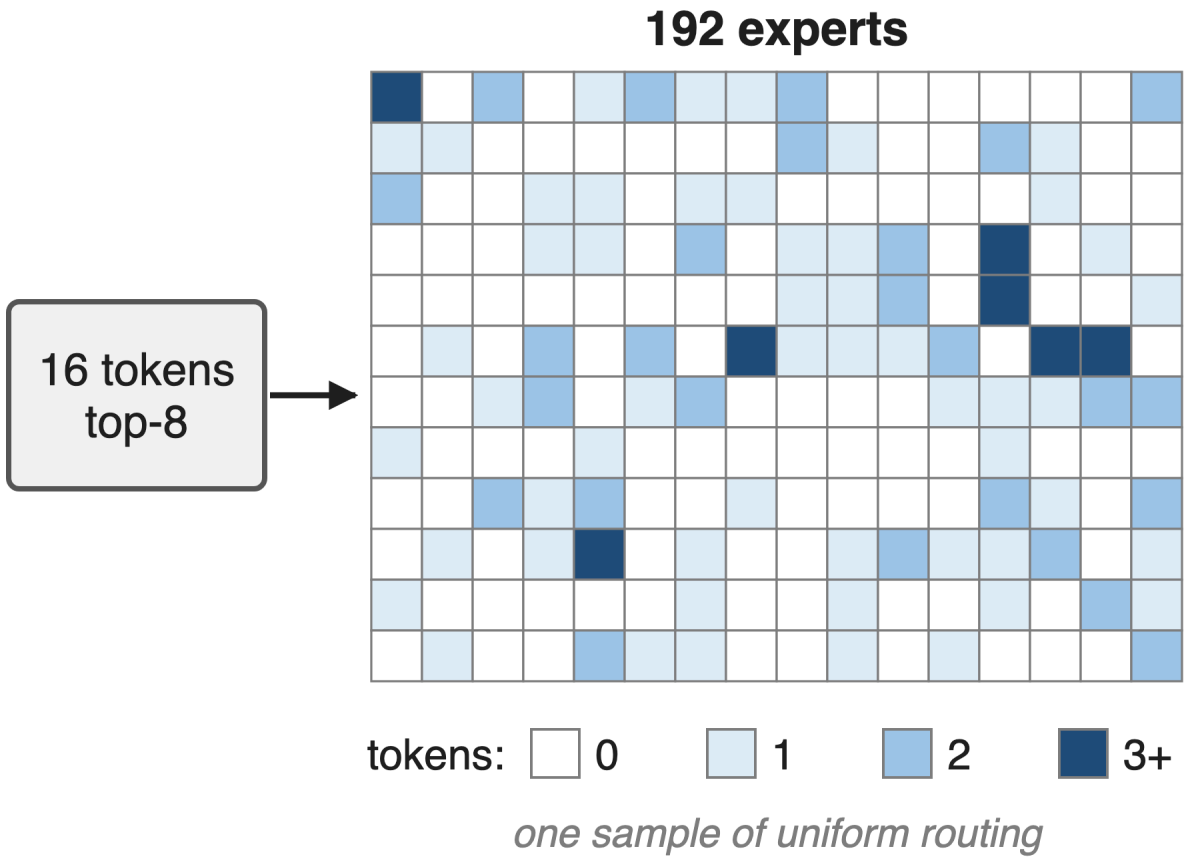
Tencent Hunyuan AI Infra · HPC-Ops

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

GEMM & MoE Kernels · github.com/Tencent/hpc-ops

Decode spreads a few tokens over many experts

- Hunyuan-V3 shape: **192** experts, top-**8**, hidden **4096**
- TP8, batch 16: **0.67** tokens per expert
- FP8 e4m3 weights and activations, H20



one sample of uniform routing

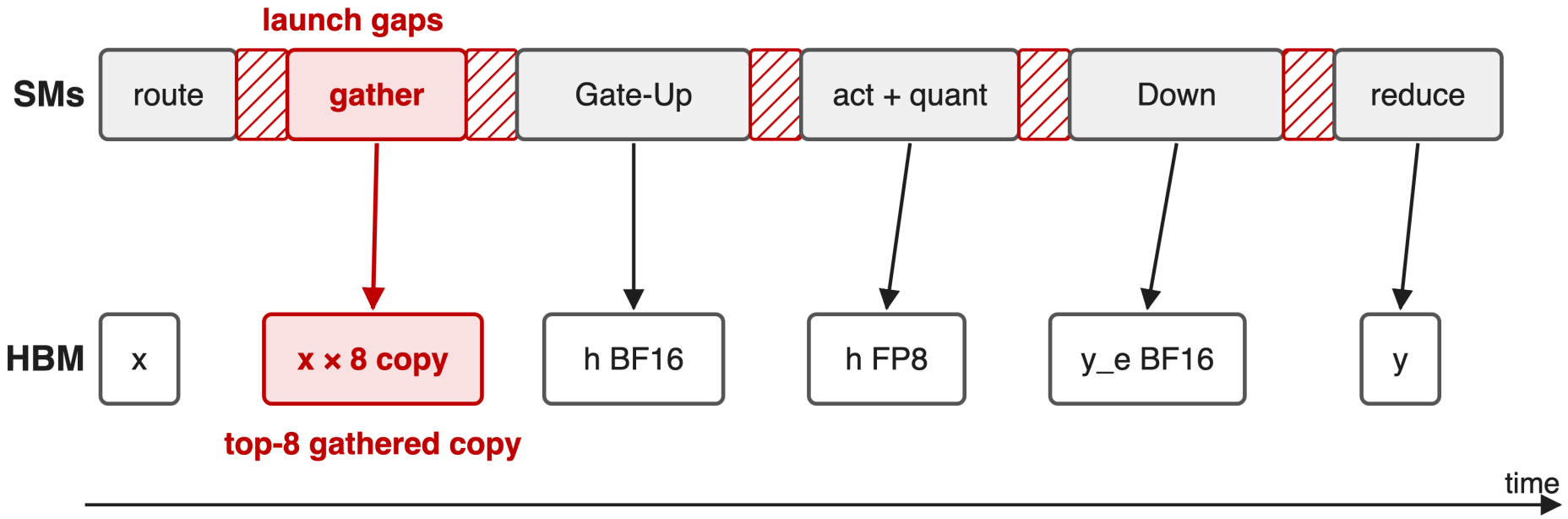
A top-8 gather copies every token 8 times

8×

token rows a gather-then-GEMM MoE writes to HBM before its first expert GEMM can start (top-8)

Gathered buffer has num_seq × top-k rows (HPC-Ops default path, src/fuse_moe/entry.cc); gather-then-GEMM design per HPC-Ops PR #49 and the LMSYS blog

Six kernels, a copy and five launch gaps



Schematic, not to scale

Routing pays a global atomic per pair; every later stage waits for a copy or a launch

Insight: pass indices, not copies

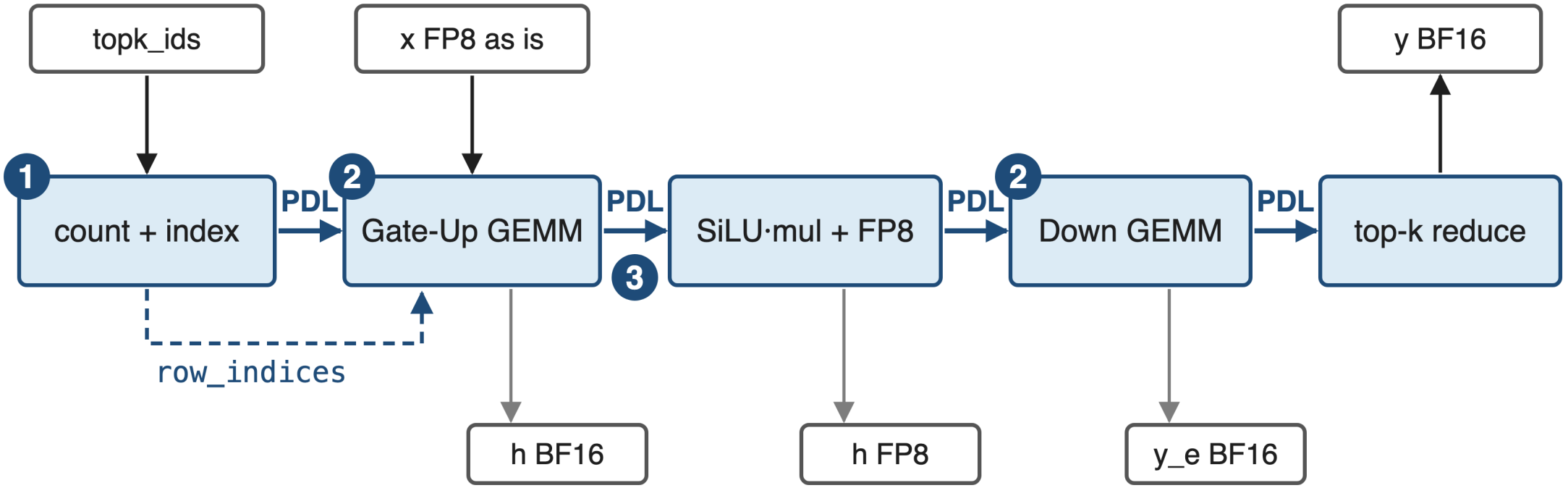
- Routing already fixes every token's slot in its expert
- An 8-row tile leaves room for many CTAs per SM

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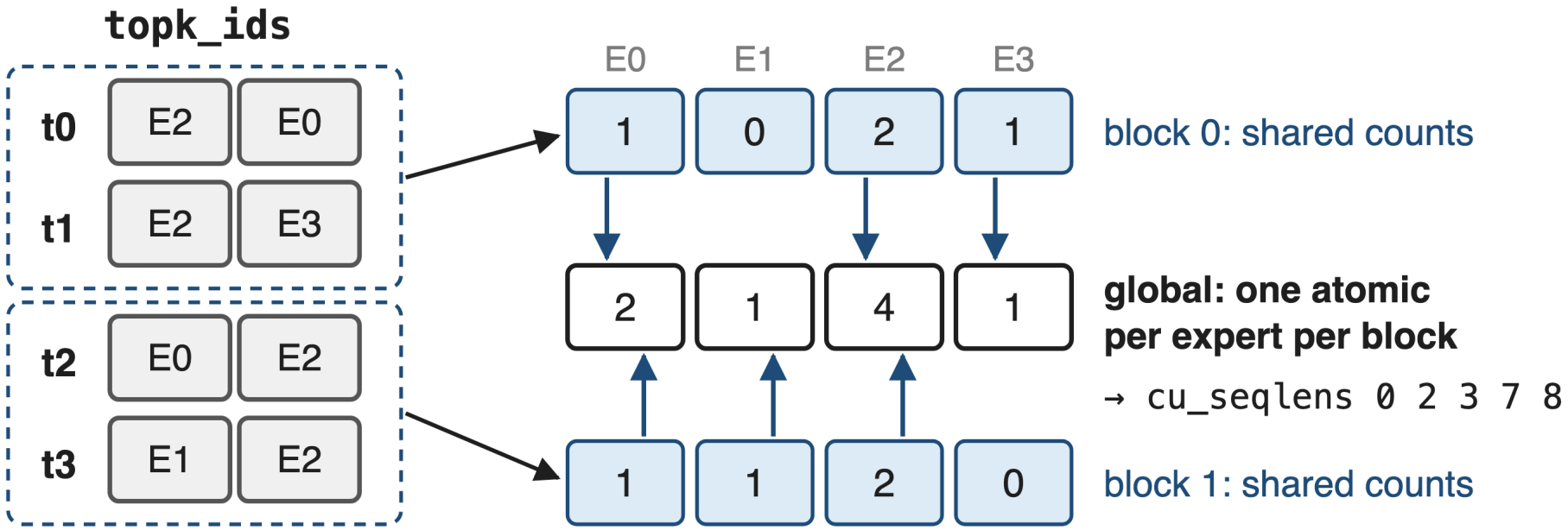
Read tokens through indices, hide latency across CTAs, and butt the stages together

Five stages, chained with PDL

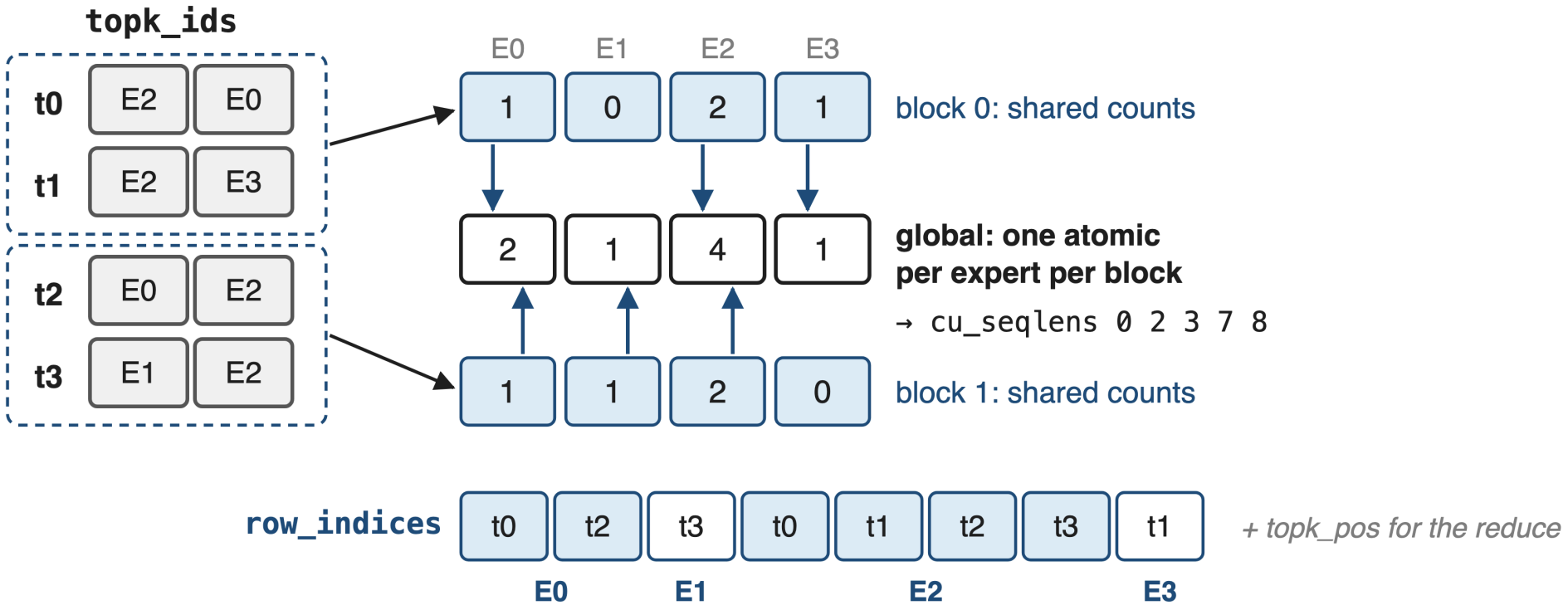


Path taken for per-tensor FP8 when the per-rank intermediate size is ≤ 512 and a multiple of 64

❶ Count in shared memory, reserve once

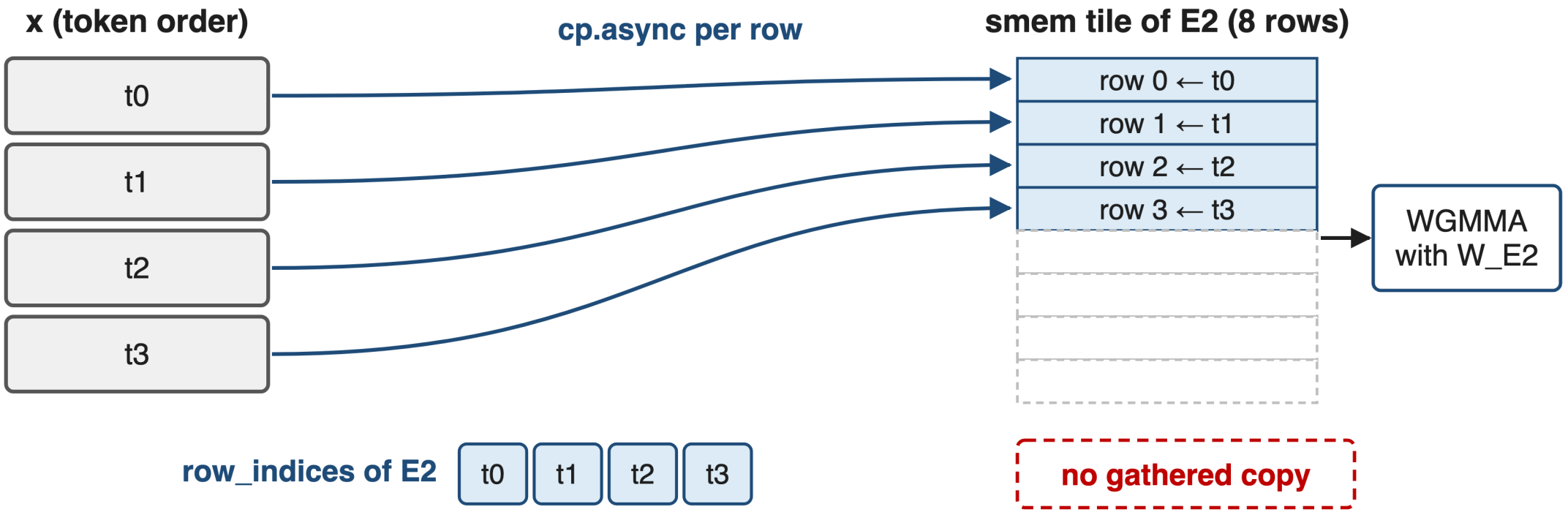


❶ Count in shared memory, reserve once



row_indices tells the Gate-Up GEMM which token each tile row is

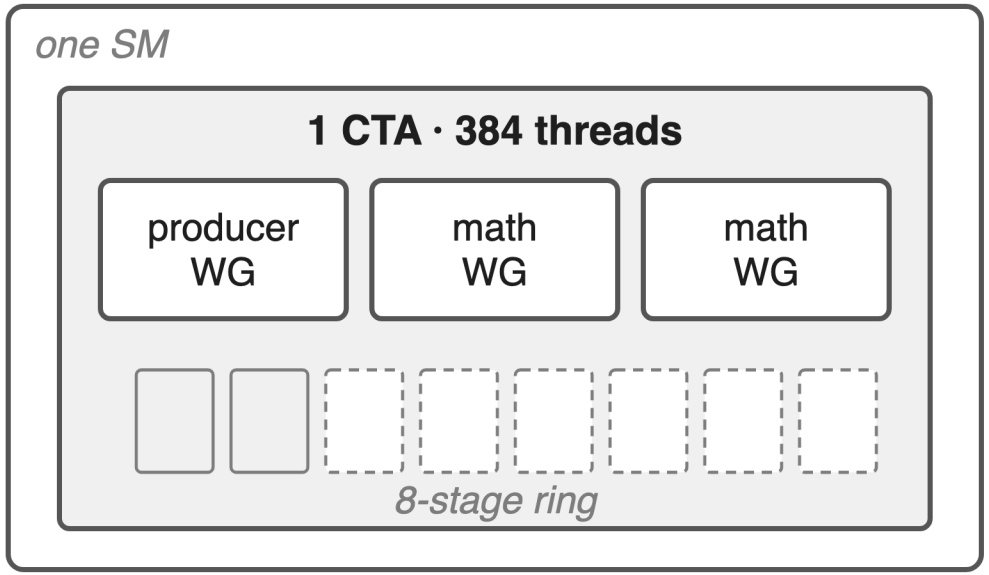
❶ Gate-Up reads tokens in place



The gathered copy and its kernel disappear; x is read once per routed row

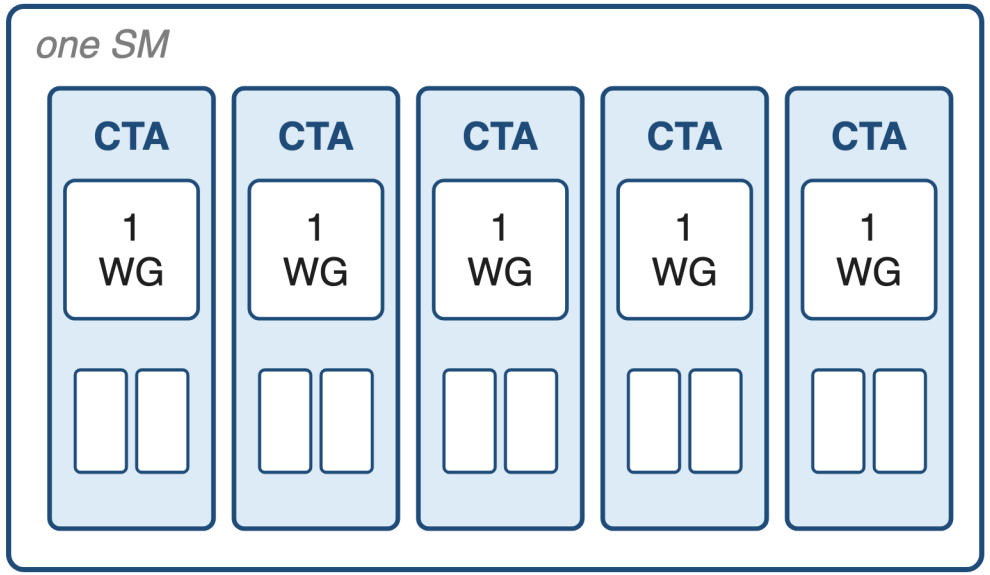
❷ Many small CTAs beat one deep pipeline

TMA + warp specialization



1 CTA per SM · 8 stages

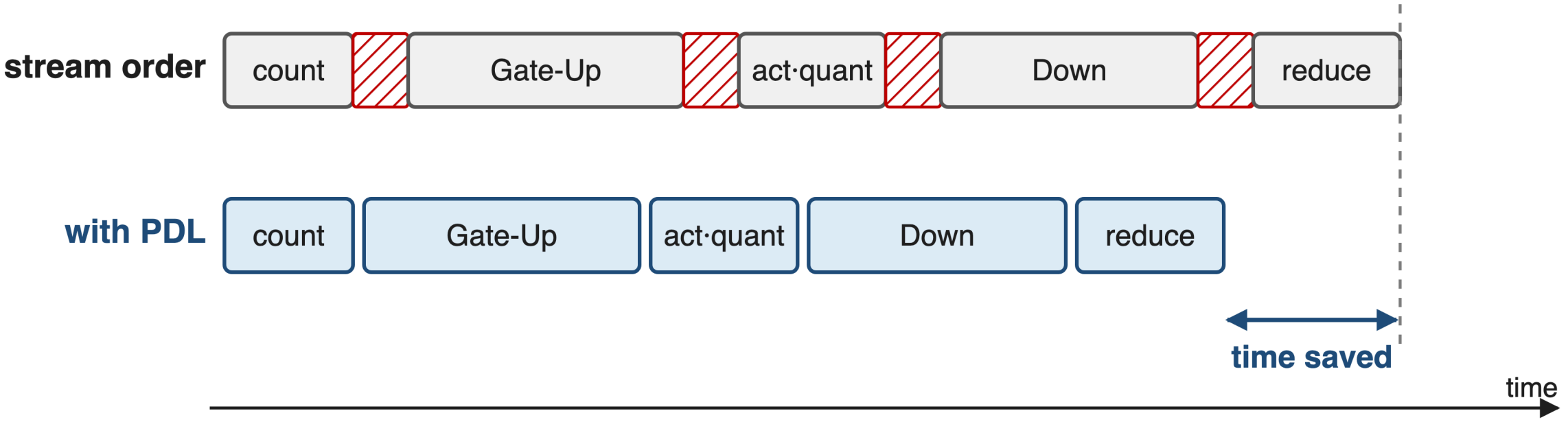
cp.async, no specialization



≥ 5 CTAs per SM · 2 stages each

Latency hiding moves from inside one CTA to the SM's warp scheduler

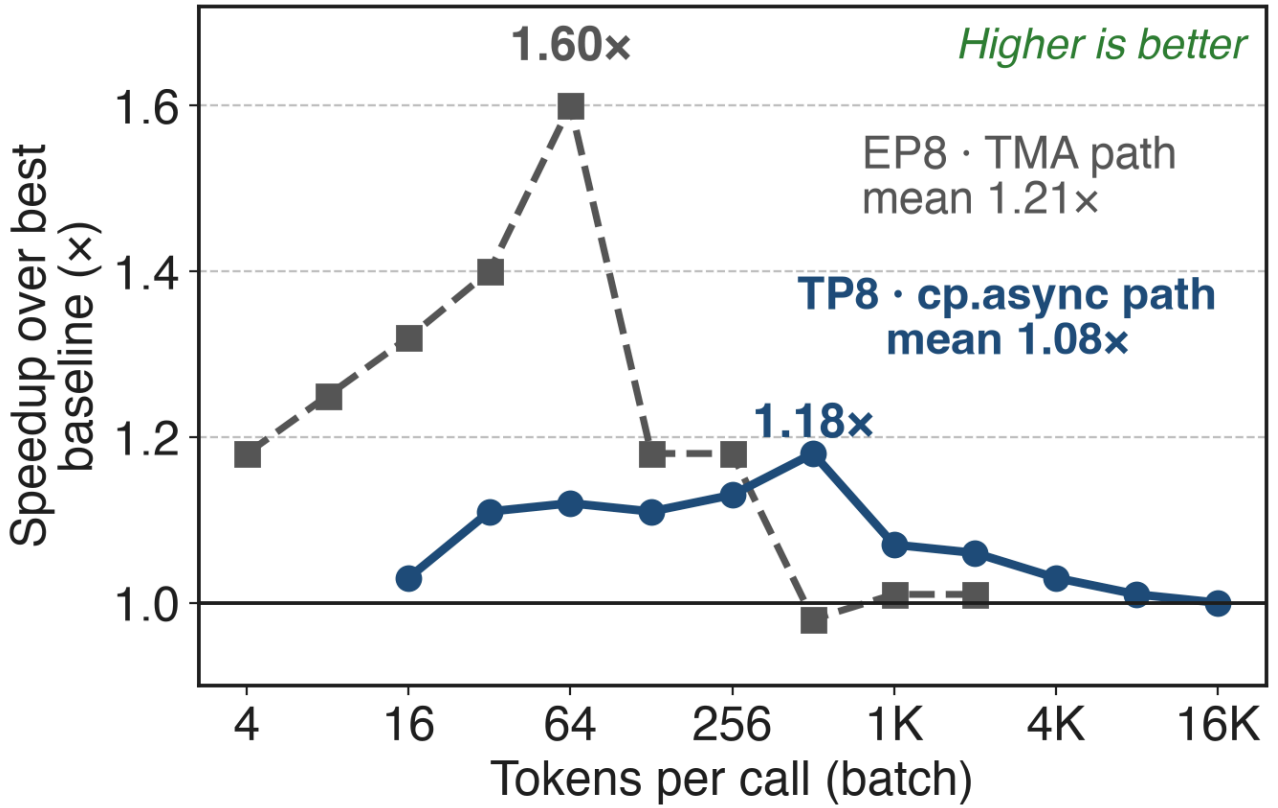
❸ PDL hides the gaps between stages



Schematic, not to scale

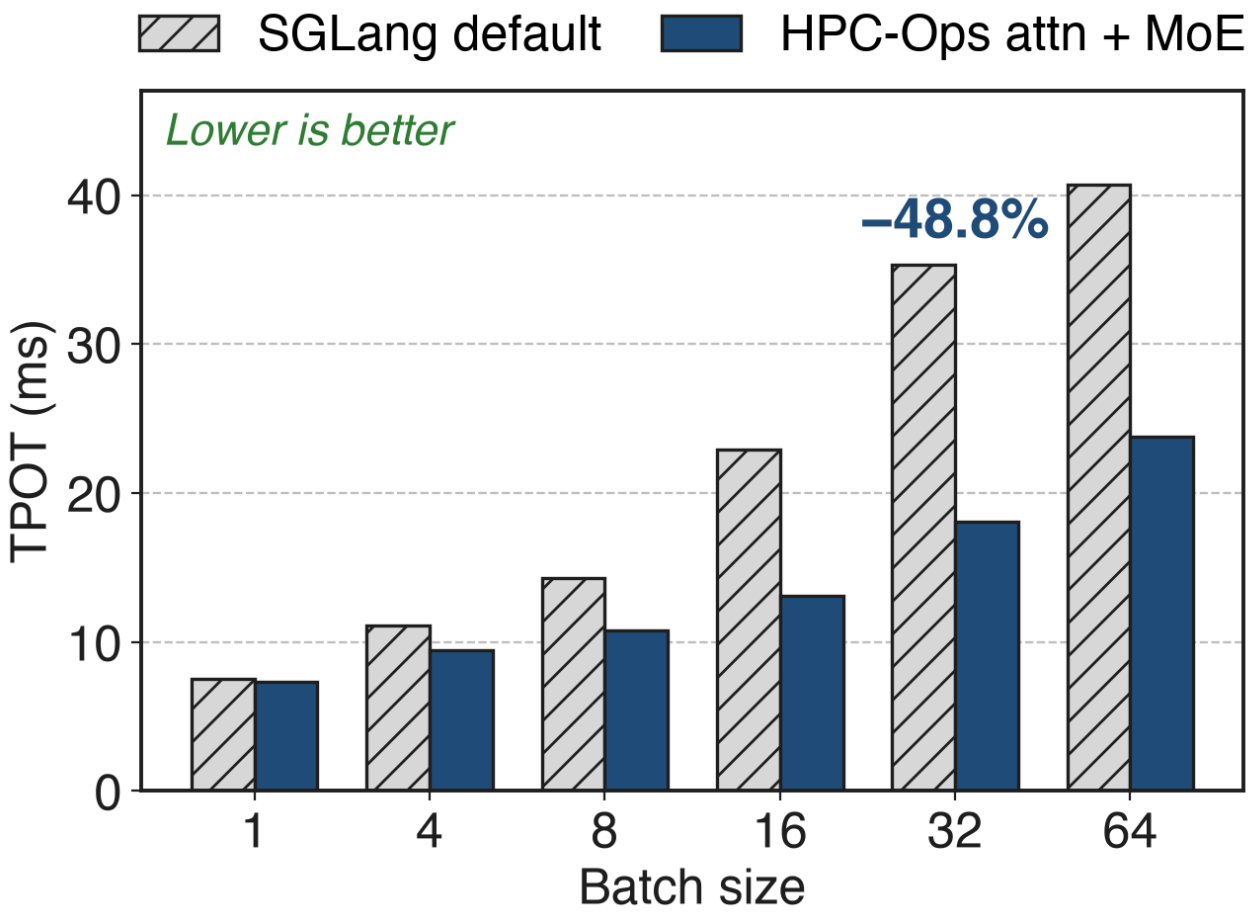
1.08× mean over the best baseline at TP8

- Hy3 shapes, H20, best baseline per batch
- TP8: up to **1.18×**, never slower
- EP8 (TMA path): up to **1.60×**



Hy3 TPOT drops up to 48.8% end-to-end

- 8×H20, TP8, FP8 KV cache
- 8K input, 4K output
- Attention + MoE together



The price: narrow fast path, honest ceilings

- cp.async path: per-tensor FP8, intermediate/rank ≤ **512**
- Wider experts, block-wise: TMA path
- Intermediates still round-trip HBM

- Not always faster: **0.98×** (EP8, 512)
- H200 Qwen3 kernel: **0.89–4.21×**
- H200 Hy3 serving: **–4.2%** to **+6.3%**

Summary

- Decode MoE: **0.67** tokens per expert; a top-8 gather copies each token **8×**
- **Insight**: pass indices, not copies; many small CTAs; PDL between stages
- **1.08×** (TP8) and **1.21×** (EP8) mean over the best baseline; up to **1.60×**
- Hy3 TPOT **–48.8%** with HPC-Ops attention + MoE

https://github.com/Tencent/hpc-ops